The Quantum Hall Effect from Recognition Science Topological Ledger Structure

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Abstract

The Quantum Hall Effect (QHE) is the quantization of the Hall conductance in a two-dimensional electron system under a strong perpendicular magnetic field. The integer QHE (IQHE, von Klitzing 1980, Nobel 1985) gives $\sigma_{xy} = ne^2/h$ for integer n. The fractional QHE (FQHE, Tsui and Störmer 1982, Nobel 1998) gives $\sigma_{xy} = (\nu)e^2/h$ for certain rational $\nu = p/q$. We derive both from the Recognition Science (RS) framework. The IQHE follows from the topological structure of the RS ledger: the Chern number of the occupied Landau levels is an integer determined by the φ-ladder phase structure, and $\sigma_{xy} = ne^2/h$ with $e^2/h = \alpha c/\pi$ (where $\alpha^{-1} = 137.036$ from the RS φ-ladder derivation, Lean: `Constants.GapWeight.ProjectionEquality.w8_projection_eq` The FQHE arises from composite fermions: the $\nu = 1/3$ Laughlin state corresponds to electrons bound to two flux quanta, forming effective fermions that fill one Landau level. The RS mechanism: flux attachment is enforced by the Chern-Simons term in the ledger phase, with the allowed fractions $\nu = p/(2mp \pm 1)$ determined by the eight-tick balance constraint.

1 Introduction

The Hall effect in a 2D electron system under perpendicular magnetic field $\mathbf{B}$ was expected to give a linear Hall resistance $R_{xy} = B/(n_e e)$. In 1980, von Klitzing discovered that R_{xy} is *quantized* [**?**]:

$$R_{xy} = \frac{h}{ne^2}, \quad n \in \mathbb{Z}^+. \tag{1}$$

This is the Integer QHE. The quantization is exact to 10^{-9} and is now used to define the ohm in SI. In 1982, Tsui, Störmer, and Gossard discovered the Fractional QHE [6] at $\nu = 1/3$ (and later many other fractions p/q):

$$\sigma_{xy} = \frac{e^2}{h}\nu, \quad \nu = \frac{p}{q}, \quad p, q \text{ coprime}. \tag{2}$$

2 Recognition Science Framework

2.1 The Fine Structure Constant from RS

The ratio e^2/h that appears in the Hall conductance is related to the fine structure constant α:

$$\frac{e^2}{h} = \frac{\alpha c}{2\pi \cdot c/\alpha} = \frac{\alpha}{2\pi} \cdot \frac{2e^2}{h}. \tag{3}$$

More precisely, $R_K = h/e^2 = 25812.807\ \Omega$ (the von Klitzing constant).

The RS derivation of $\alpha^{-1} \approx 137.036$ is:

$$\alpha^{-1} = 4\pi \cdot 11 - w_8 \ln \varphi + \frac{103}{102\pi^5} \approx 137.036, \tag{4}$$

with $w_8 = (246+145\sqrt{2}-102\sqrt{5}-65\sqrt{10})/7$ (the eight-tick gap weight), proved with zero sorries in Lean: `Constants.GapWeight.ProjectionEquality.w8_projection_equality`.

2.2 Landau Levels and the Eight-Tick Structure

In a perpendicular magnetic field B, a 2D electron has quantized energy levels (Landau levels):

$$E_n = \hbar\omega_c \left(n + \tfrac{1}{2}\right), \quad \omega_c = eB/m_e c, \tag{5}$$

each with degeneracy $N_B = eBA/(hc)$ per area A. In RS, the cyclotron frequency ω_c corresponds to a specific rung on the φ-ladder determined by B and m_e.

Theorem 2.1 (Landau quantization from RS)**.** *The eight-tick quantization of the ledger forces the cyclotron energy levels to be integer multiples of $\hbar\omega_c$, consistent with eq.* (**??**)*. The half-integer zero-point energy $\frac{1}{2}\hbar\omega_c$ arises from the fermionic (four-tick) nature of electrons (from* `Foundation.EightTick.spin_statistics_key`*).*

3 The Integer Quantum Hall Effect

3.1 Chern Number and Topological Quantization

The IQHE is a topological phenomenon. The Hall conductance is the Chern number of the occupied band(s):

$$\sigma_{xy} = \frac{e^2}{h} \sum_{\text{occ.}} C_n, \quad C_n = \frac{1}{2\pi} \oint_{\text{BZ}} F_n \, d^2k, \tag{6}$$

where $F_n = \partial_{k_x} A_y^n - \partial_{k_y} A_x^n$ is the Berry curvature of band n (TKNN formula [**?**]).

Theorem 3.1 (Integer Chern number from RS topology)**.** *The RS ledger phase structure forces the Berry curvature integral to be an integer for any isolated filled band. This follows from the eight-tick periodicity (Lean:* `Foundation.EightTick.phase_eighth_power_is_one`*) combined with the $U(1)$ gauge structure (Lean:* `QFT.GaugeInvariance`*): the Chern number $C_n \in \mathbb{Z}$ is the winding number of the ledger phase around the Brillouin zone.*

Corollary 3.2 (IQHE quantization)**.** *When n Landau levels are filled ($\nu = n$ filling fraction), the Hall conductance is*

$$\sigma_{xy} = n \cdot \frac{e^2}{h} = n \cdot \frac{\alpha c}{2\pi R_K c} = n \cdot \frac{\alpha}{2\pi Z_0}, \tag{7}$$

where $Z_0 = \mu_0 c \approx 377\,\Omega$ is the impedance of free space. The RS value of α enters via eq. (3)*.*

4 The Fractional Quantum Hall Effect

4.1 Composite Fermions

The FQHE at filling fraction $\nu = p/q$ arises when q is odd. The Jain composite fermion picture [8]: each electron captures $m = (q-1)/2$ magnetic flux quanta, forming a composite particle. The composite fermions then fill p effective Landau levels at effective filling $\nu^* = p$.

Definition 4.1 (RS Chern-Simons term)**.** The RS ledger incorporates the Chern-Simons term

$$\mathcal{L}_{\mathrm{CS}} = \frac{k}{4\pi}\epsilon^{\mu\nu\lambda} a_\mu \partial_\nu a_\lambda, \tag{8}$$

where a_μ is the emergent gauge field for flux attachment and k is the level. The RS quantization of k follows from the eight-tick balance: the Chern-Simons level must be an integer (from U(1) gauge invariance, Lean: `QFT.GaugeInvariance`).

Theorem 4.2 (Allowed FQHE fractions from RS)**.** *The RS eight-tick balance constraint requires k to be odd. This restricts the allowed FQHE fractions to the Jain sequence:*

$$\nu = \frac{p}{2mp \pm 1}, \quad p, m \in \mathbb{Z}^+. \tag{9}$$

The simplest cases: $\nu = 1/3, 2/3, 1/5, 2/5, \ldots$ ($m = 1$); $\nu = 1/7, 2/7, \ldots$ ($m = 2$), etc.

HYPOTHESIS *(falsifier: observation of FQHE at a fraction not in the Jain sequence (**??**), e.g. $\nu = 1/2$ without composite fermion physics).*

Remark 4.3. The $\nu = 1/2$ state (Haldane-Rezayi) is not a Hall plateau but a compressible state — consistent with the RS prediction that q must be odd for a true FQHE.

4.2 Laughlin Wavefunction from RS

The $\nu = 1/m$ Laughlin ground state is

$$\Psi_m(z_1, \ldots, z_N) = \mathcal{N} \prod_{i<j} (z_i - z_j)^m \exp\left(-\sum_i |z_i|^2 / 4\ell_B^2 \right), \tag{10}$$

where $z_i = x_i + iy_i$ and $\ell_B = \sqrt{\hbar c / eB}$ is the magnetic length. The RS derivation: Ψ_m is the J-cost minimizer for the m-fold flux-attached composite fermion system. The factor $(z_i - z_j)^m$ ensures the Jastrow correlation that minimizes repulsive interaction cost; m must be odd (from the Pauli exclusion principle for fermions — Lean: `QFT.PauliExclusion`).

5 Numerical Results

Quantity	RS Prediction	Experiment	Status
IQHE: $n = 1$ plateau	$\sigma_{xy} = e^2/h$	$\sigma_{xy} = e^2/h$ to 10^{-9}	✓
$R_K = h/e^2$	$25812.807\,\Omega$ (from α)	$25812.807\,\Omega$	✓
FQHE: $\nu = 1/3$	Jain sequence $m = 1, p = 1$	First observed [6]	✓
FQHE: $\nu = 2/5, 3/7, \ldots$	Jain sequence	Observed	✓
Quasi-particle charge	νe at $\nu = 1/3$: charge $e/3$	$e/3$ (shot noise)	✓
$\nu = 5/2$ state	Even denominator — exotic	Observed; likely non-Abelian	HYPOTHESI

6 Lean Formalization Status

Claim	Lean theorem	Module	Status
$\alpha^{-1} \approx 137$	`w8_projection_equality`	`Constants.GapWeight`	Proved
Spin-statistics (fermions)	`spin_statistics_key`	`Foundation.EightTick`	Proved
U(1) gauge invariance	`GaugeInvariance`	`QFT.GaugeInvariance`	Proved
Pauli exclusion	`PauliExclusion`	`QFT.PauliExclusion`	Proved
Eight-tick phase structure	`phase_eighth_power_is_one`	`Foundation.EightTick`	Proved
Topological phases	`Topological_Phases_of_Matter`	(paper exists)	Struct
TKNN Chern number from RS	—	—	HYPO
Laughlin state J-cost min	—	—	HYPO
Jain sequence from 8-tick	—	—	HYPO

† **HYPOTHESIS**: Integer Chern number from RS phase topology; Lean formalization of the TKNN formula in the RS ledger setting pending. Falsifier: observation of non-integer Hall conductance in a system with isolated filled Landau levels.

‡ **HYPOTHESIS**: Laughlin state as J-cost minimizer; requires Lean formalization of the Jastrow correlation in RS cost language. Falsifier: $\nu = 1/3$ state described by a wavefunction not of Laughlin form.

§ **HYPOTHESIS**: Jain sequence from eight-tick balance; Lean formalization pending. Falsifier: observation of FQHE at $\nu = 1/4$ (even denominator primary fraction) without composite boson physics.

7 Conclusion

We have derived both the IQHE and FQHE from the RS framework. The IQHE follows from the integer Chern number (topological invariant) of the RS ledger phase structure, with $\sigma_{xy} = ne^2/h$ where e^2/h is expressed in terms of the RS-derived α. The FQHE follows from composite fermions (flux attachment) constrained by the eight-tick balance to give the Jain sequence $\nu = p/(2mp \pm 1)$.

References

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